Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

Learning goal: I can convert units using proportional reasoning. (circle your level after scoring)

4	I can solve a multi-step conversion problem across measurement systems and check the result.
3	I can convert between measurement systems for length, area, weight or mass, volume, or money, using a conversion factor.
2	I can set up a ratio between two units.
1	I need help to start this benchmark.

1.

$1 \text{ kilogram} \approx 2.2 \text{ pounds}.$

A suitcase weighs 40 pounds. What is its mass in kilograms, to the nearest tenth?

A. 1.8 kilograms

B. 37.8 kilograms

C. 18.2 kilograms

D. 88.0 kilograms

2. On one day, 1 U.S. dollar could be exchanged for 0.9 Swiss francs.

Maya exchanged $100.00 for Swiss francs and then spent 31.25 Swiss francs on a meal. How many Swiss francs did Maya have left?

Turn over →

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____

3.

$1 \text{ mile} \approx 1.609 \text{ kilometers}$.

A race course is 36 kilometers long. How many miles is that? Round to the nearest tenth.

4. $1 \text{ hour} = 60 \text{ minutes}$, and $1 \text{ gallon} \approx 3.785 \text{ liters}$.

A pump moves 20 liters of water each minute. It runs for 3 hours 15 minutes. How many gallons does it move? Round to the nearest tenth.

5. Store A sells cheddar cheese for $7.29 per pound. Store B sells cheddar cheese for $17.05 per kilogram.

$1 \text{ kilogram} \approx 2.2 \text{ pounds}$.

Which store charges less per pound, and how much less per pound?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.3.3 · MA.7.AR.3.3 · Unit conversions across measurement systems

Unit 6: Multi-Step Problems With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **C. 18.2 kilograms**

Each kilogram is about 2.2 pounds, so there are fewer kilograms than pounds: divide by 2.2.

$40 \div 2.2 \approx 18.182$, which rounds to 18.2 kilograms.

2. Regular **58.75 Swiss francs**

Each dollar becomes 0.9 Swiss francs: $100 \times 0.9 = 90.00$ Swiss francs.

Subtract the meal: $90.00 - 31.25 = 58.75$ Swiss francs.

3. Regular **22.4 miles**

Each mile is about 1.609 kilometers, so divide the kilometers by 1.609: $36 \div 1.609 \approx 22.374$.

Rounded to the nearest tenth: 22.4 miles.

There are fewer miles than kilometers, so the answer must be smaller than 36.

4. Regular **1030.4 gallons**

Convert the time to minutes: $3 \times 60 + 15 = 195$ minutes.

Liters moved: $20 \times 195 = 3900$ liters.

Liters to gallons: $3900 \div 3.785 \approx 1030.383$, which rounds to 1030.4 gallons.

5. Rigor **Store A, by $0.46 per pound. Score: 1 point for the store and the amount per pound; the store alone earns 0.**

Store B's price is for about 2.2 pounds, so divide by 2.2 to get the price per pound: $\$17.05 \div 2.2 = \$7.75$ per pound.

Store A charges $7.29 per pound.

$\$7.75 - \$7.29 = \$0.46$, so Store A charges $0.46 less per pound.

Re-check strip (3 items, C-AR.3.3-R)

1. Regular **D. 26.4 kilograms**

Each kilogram is about 2.2 pounds, so there are fewer kilograms than pounds: divide by 2.2.

$58 \div 2.2 \approx 26.364$, which rounds to 26.4 kilograms.

2. Regular **69.75 euros**

Each dollar becomes 0.88 euros: $100 \times 0.88 = 88.00$ euros.

Subtract the meal: $88.00 - 18.25 = 69.75$ euros.

3. Regular **30.6 gallons**

Each gallon is about 3.785 liters, so divide the liters by 3.785: $116 \div 3.785 \approx 30.647$.

Rounded to the nearest tenth: 30.6 gallons.

Comparison strip (2 items, C-AR.3.3-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B. 12.7 kilograms**

Each kilogram is about 2.2 pounds, so there are fewer kilograms than pounds: divide by 2.2.

$28 \div 2.2 \approx 12.727$, which rounds to 12.7 kilograms.

2. Regular **96.5 kilometers (96.6 also accepted if the factor was rounded to 1.61)**

Convert minutes to hours: $75 \div 60 = 1.25$ hours.

Distance: $48 \times 1.25 = 60$ miles.

Miles to kilometers: $60 \times 1.609 \approx 96.540$, which rounds to 96.5 kilometers.

Reteach plan — C-AR.3.3

MA.7.AR.3.3 · Unit conversions across measurement systems

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items)

C-AR.3.3-R

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

1.

1 kilogram $\approx$ 2.2 pounds.

A suitcase weighs 58 pounds. What is its mass in kilograms, to the nearest tenth?

A. 55.8 kilograms

B. 127.6 kilograms

C. 2.6 kilograms

D. 26.4 kilograms

2. On one day, 1 U.S. dollar could be exchanged for 0.88 euros.

Maya exchanged $100.00 for euros and then spent 18.25 euros on a meal. How many euros did Maya have left?

3.

1 gallon $\approx$ 3.785 liters.

A water heater holds 116 liters. How many gallons is that? Round to the nearest tenth.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items)

C-AR.3.3-R

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

1.

1 kilogram $\approx$ 2.2 pounds.

A suitcase weighs 58 pounds. What is its mass in kilograms, to the nearest tenth?

A. 55.8 kilograms

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C. 2.6 kilograms

D. 26.4 kilograms

2. On one day, 1 U.S. dollar could be exchanged for 0.88 euros.

Maya exchanged $100.00 for euros and then spent 18.25 euros on a meal. How many euros did Maya have left?

3.

1 gallon $\approx$ 3.785 liters.

A water heater holds 116 liters. How many gallons is that? Round to the nearest tenth.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.3.3-X

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

1.

$1 \text{ kilogram} \approx 2.2 \text{ pounds}.$

A suitcase weighs 28 pounds. What is its mass in kilograms, to the nearest tenth?

A. 25.8 kilograms

B. 12.7 kilograms

C. 61.6 kilograms

D. 1.3 kilograms

2. $1 \text{ hour} = 60 \text{ minutes}$, and $1 \text{ mile} \approx 1.609 \text{ kilometers}$.

A car travels at 48 miles per hour for 75 minutes. How many kilometers does it travel? Round to the nearest tenth.

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-AR.3.3-X

Benchmark: **MA.7.AR.3.3** · Unit conversions across measurement systems Student code: _____ Calculator: yes

1.

$1 \text{ kilogram} \approx 2.2 \text{ pounds}.$

A suitcase weighs 28 pounds. What is its mass in kilograms, to the nearest tenth?

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2. $1 \text{ hour} = 60 \text{ minutes}$, and $1 \text{ mile} \approx 1.609 \text{ kilometers}$.

A car travels at 48 miles per hour for 75 minutes. How many kilometers does it travel? Round to the nearest tenth.

Score: ____ / 2 → **copy to your progress strip**